

Setting Up an Amazon RDS MySQL Database and Connecting To It

To Begin with the Lab

Summary of the Lab

In this lab, an **Amazon RDS** MySQL database was created using the AWS Management Console with a minimal free-tier setup. The database engine version was selected, a database name, admin username, and password were configured, and public access was enabled so the database could be reached from a local SQL client. After the instance finished provisioning, the endpoint was copied and used in **MySQL Workbench** to establish a connection. A new database was then created from the client side to show how database administration is handled after the RDS instance is available. Finally, the setup was verified by successfully connecting to the database and confirming that the instance could be managed from the SQL client.

Understanding Amazon RDS and SQL Client Access

Amazon RDS is a managed database service that lets you create and run databases without handling the low-level server setup yourself. A SQL client such as **MySQL Workbench** is used to connect to the database, run SQL commands, and manage database objects like schemas and tables. This exists because cloud engineers usually create the database infrastructure, while database users or programmers handle the actual data design and SQL operations. Public access is convenient for lab practice, but in real projects the database is usually placed in a private network and accessed through a controlled application server or bastion host.

Example:

A cloud engineer creates a MySQL RDS instance named mydb with username admin and gives the endpoint to a database developer. The developer opens **MySQL Workbench**, connects using the RDS endpoint, and creates a new application database called myappdb. This separates infrastructure setup from database usage and makes administration easier.

Lab Steps

- Sign in to the AWS Management Console.
- Open **Aurora and RDS**.
- Choose **Create with full configuration** and Click **Create database**.

Welcome to Aurora and RDS [Info](#)

Amazon Relational Database Service (Amazon RDS) and Amazon Aurora are fully managed database services that simplify setup, operation, and scaling of relational databases in the cloud.



Create with express configuration in seconds

Create and query an Aurora PostgreSQL serverless database with pre-configured settings to get started quickly.

[Create](#)



Create with full configuration

Create a database with custom configurations and enable features you want across a variety of engines including Aurora (compatible with MySQL & PostgreSQL), MySQL, PostgreSQL, MariaDB, Oracle, Microsoft SQL Server, and IBM Db2. Or use an existing database using restore from S3.

[Create](#) [Restore from S3](#)

Service overview

Viewing data from Asia Pacific (Mumbai) (ap-south-1) Region.

DB clusters 0 **DB instances** 0 **Snapshots** 5 **Recent events** 0

[View service quota](#)

Explore Aurora & RDS

In this activity, you will learn how to create a database. To begin, choose **Start tutorial**.

Estimated duration 2-5 minutes

[Start tutorial](#)

Service health

Current status [Details](#) [View service health dashboard](#)

RDS costs [View](#)

- Choose your **Availability Zone**.
- Select **MySQL** as the database engine.

Create database [Info](#)

Availability zone group [Info](#)

Availability zone group

Select the availability zone where you want to create your database.

Asia Pacific (Mumbai)

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible) ☐ Aurora (PostgreSQL Compatible) ☒ **MySQL** ☐ PostgreSQL

☐ MariaDB ☐ Oracle ☐ Microsoft SQL Server ☐ IBM Db2

Choose a database creation method

☒ **Full configuration** ☐ Easy create

- Choose database creation method as **Full configuration**.
- Choose the **Free tier** template.

Choose a database creation method

☒ **Full configuration**
Choose an option that provides configuration options, including ones for availability, security, backups, and maintenance.

☐ **Easy create**
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Templates
Choose a sample template to meet your use case.

☐ **Production**
Use defaults for high availability and fast, consistent performance.

☐ **Dev/Test**
This instance is intended for development use outside of a production environment.

☒ **Free tier**
Use the Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS. [Info](#)

Availability and durability
Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. [Learn more in the Amazon RDS service level agreement \(SLA\)](#)

Deployment options

- ☐ **Multi-AZ DB cluster deployment (3 instances)**
Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:
 - 99.95% uptime
 - Redundancy across Availability Zones
 - Increased read capacity
 - Reduced write latency
- ☐ **Multi-AZ DB instance deployment (2 instances)**
Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:
 - 99.95% uptime
 - Redundancy across Availability Zones
- ☒ **Single-AZ DB instance deployment (1 instance)**
Creates a primary DB instance. This setup provides:
 - 99.5% uptime
 - No data redundancy

Diagram illustrating the deployment options:

- Multi-AZ DB cluster deployment (3 instances):** Shows a primary instance in AZ 1 and two standby instances in AZ 2 and AZ 3. Endpoints include Write/read endpoint (AZ 1), Reader endpoints (AZ 2, AZ 3), and Standby (no endpoint).
- Multi-AZ DB instance deployment (2 instances):** Shows a primary instance in AZ 1 and a standby instance in AZ 2. Endpoints include Write/read endpoint (AZ 1) and Standby (no endpoint).
- Single-AZ DB instance deployment (1 instance):** Shows a single primary instance in AZ 1. Endpoint includes Write/read endpoint (AZ 1).

- Enter the database name.
- Keep the engine version at the default MySQL 8.0.x option or choose one you have installed on your computer.
- Set the master username to **admin**.

Settings

Edition
☒ **MySQL Community**

Engine version
View the engine versions that support the following database features.

Hide filters

☐ **Show only versions that support the Multi-AZ DB cluster**
Create a Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.

☐ **Show only versions that support the Amazon RDS Optimized Writes**
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

Engine version
MySQL 8.0.46

☐ **Enable RDS Extended Support**
Amazon RDS Extended Support is a paid offering. By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the [RDS for MySQL documentation](#).

DB instance identifier
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.
rd-database-1

Credentials Settings

Master username
Type a login ID for the master user of your DB instance.
admin

- Credentials manager as **Self-Managed**.
- Create and confirm a password.

Credentials Settings

Master username [Info](#)
Type a login ID for the master user of your DB instance.
admin
1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.
☐ Managed in AWS Secrets Manager - most secure
 Amazon RDS can generate a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
☒ Self managed
 Create your own password or have RDS create a password that you manage.
☐ Auto generate password
 Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

 Password strength: Weak
 Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' * @

Confirm master password [Info](#)

Additional credentials settings

Database authentication options [Info](#)
☒ Password authentication
 Authenticates using database passwords.
☐ Password and IAM database authentication
 Authenticates using the database password and user credentials through AWS IAM users and roles.

- Leave storage at the default value.
- Select the default VPC settings.

Public access [Info](#)
☒ Yes
 RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
☐ No
 RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.
☒ Choose existing
☐ Create new
 Create new VPC security group

Existing VPC security groups
Choose one or more options
 default X

Availability Zone [Info](#)
ap-south-1a

RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.
☐ Create an RDS Proxy [Info](#)
 RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional [Info](#)
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.
 rds-ca-rsa2048-g1 (default)
 Expiry: May 20, 2061

- Keep the default instance class provided by the free-tier setup.

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ **Hide filters**

☐ Show instance classes that support Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

☐ Include previous generation classes

- ☐ Standard classes (includes m classes)
- ☐ Memory optimized classes (includes r and x classes)
- ☒ Burstable classes (includes t classes)

Instance type

db.t3.micro

2 vCPUs, 2 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Storage

Storage type [Info](#)

Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp2)
Baseline performance determined by volume size

Allocated storage [Info](#)

20 GiB

Allocated storage value must be 20 GiB to 6,144 GiB

- Enable public access so the database can be reached from the local machine.
- Keep the default security group selection.
- Make sure to enable **Public Access**.

Connectivity [Info](#)

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.

Network type [Info](#)

To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ IPv4
Your resources can communicate only over the IPv4 addressing protocol.

☐ Dual-stack mode
Your resources can communicate over IPv4, IPv6, or both.

Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-072c4acd761c3b942)

2 subnets, 2 availability zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

ⓘ After a database is created, you can't change its VPC.

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

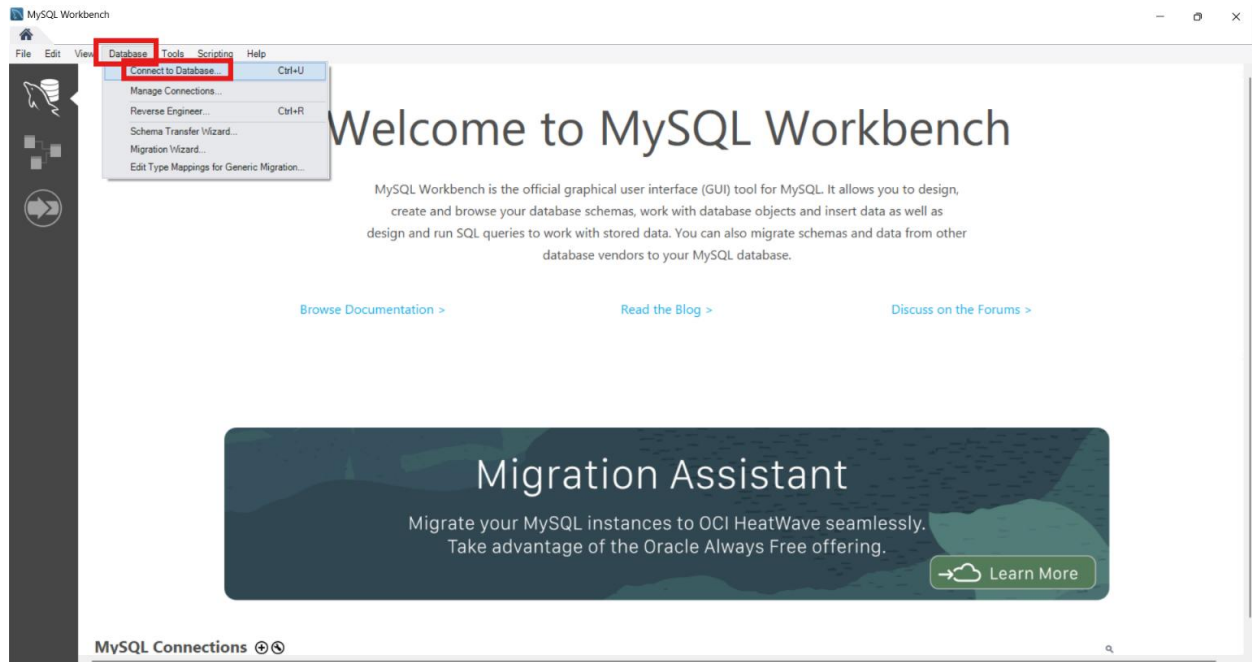
default

2 subnets, 2 Availability Zones

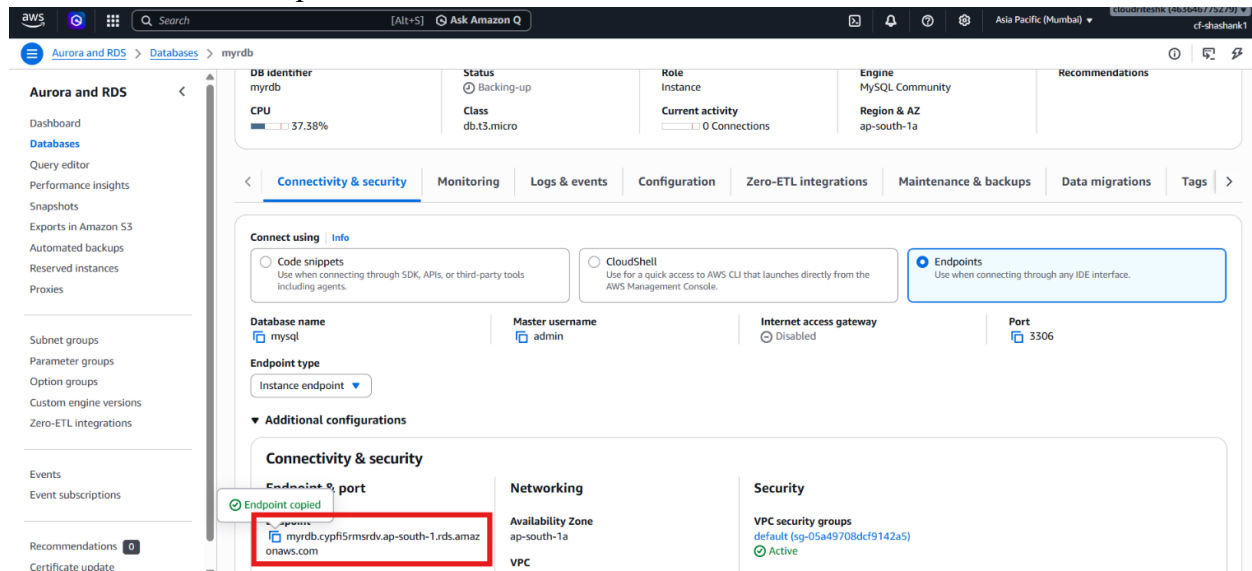
Public access [Info](#)

☒ Yes
Add a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

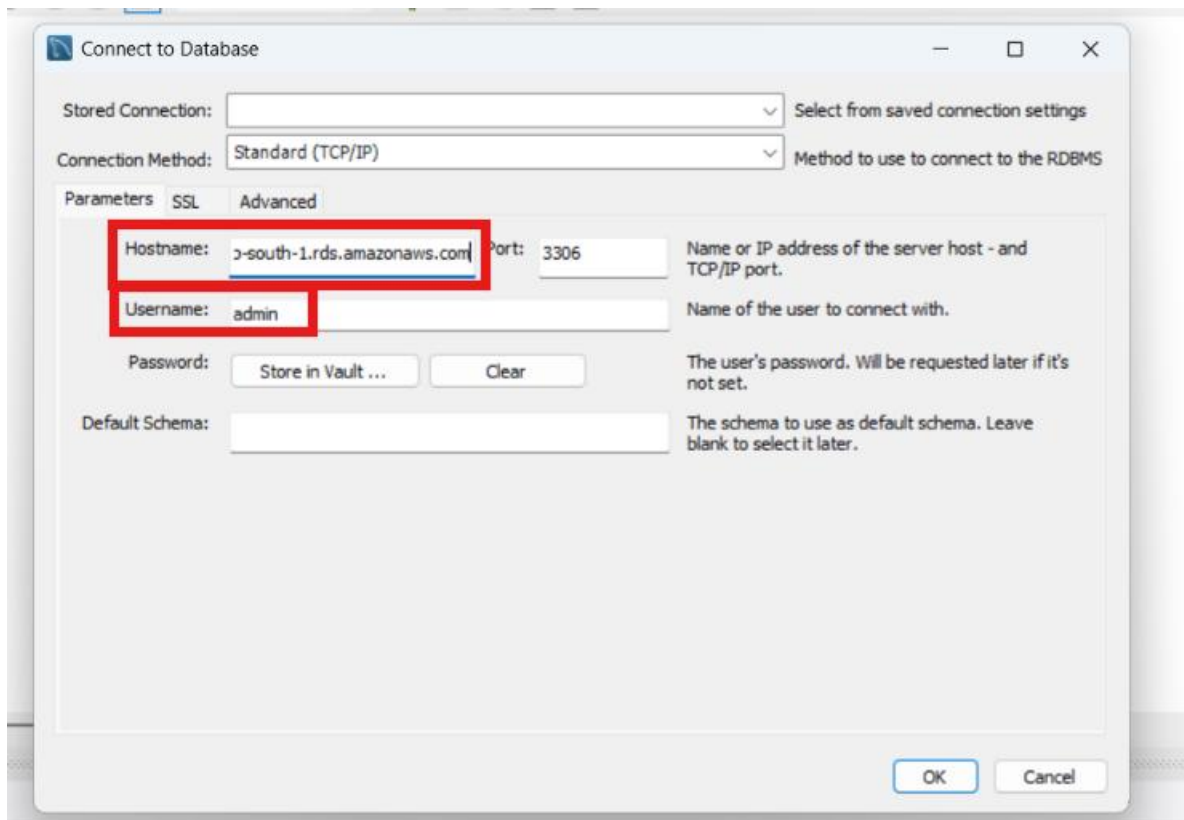
- Click **Create database**.
- Wait until the database status becomes **Available**.
- Copy the RDS endpoint from the database details page.
- Open **MySQL Workbench** on the local computer.
- Create a new MySQL connection.



- Paste the RDS endpoint into the host field.



- Use **admin** as the username.
- Test the connection.

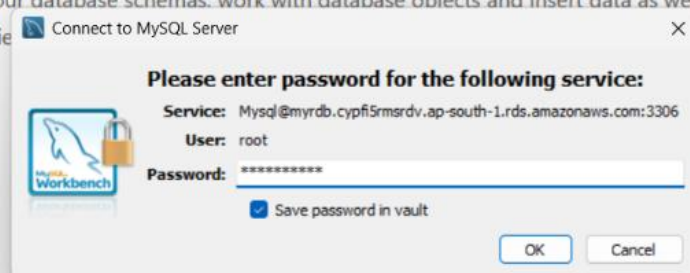


- Enter the password when prompted.
- Save the password if needed.

Welcome to MySQL Workbench

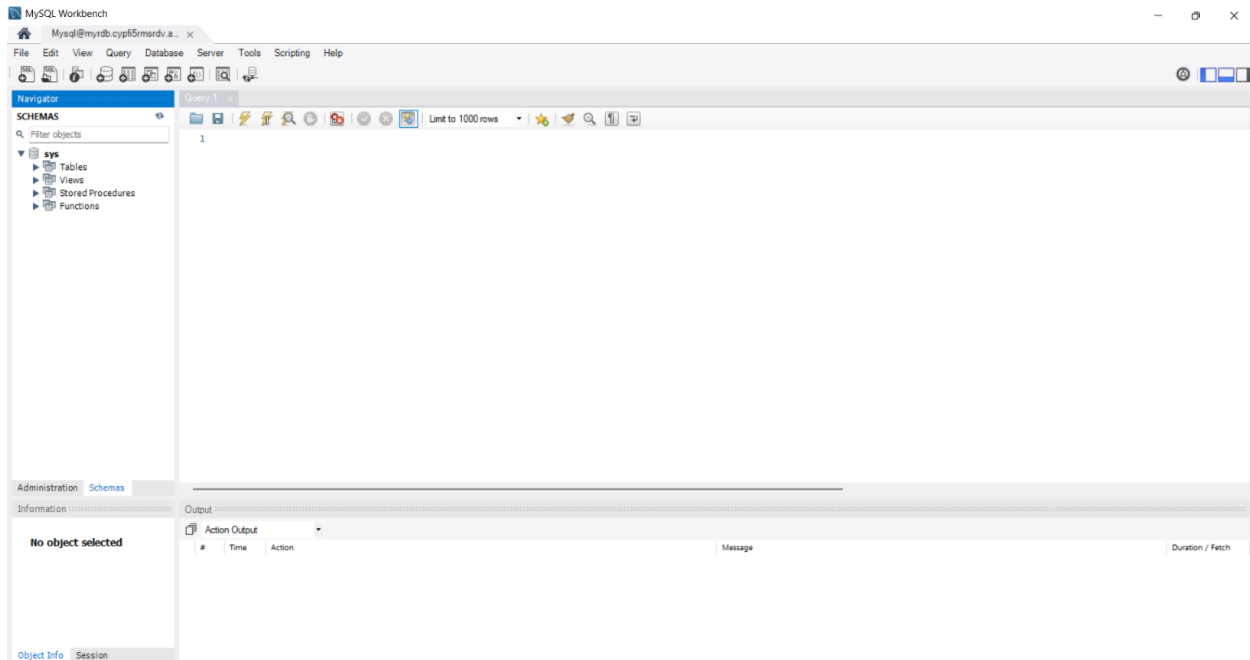
MySQL Workbench is the official graphical user interface (GUI) tool for MySQL. It allows you to design, create and browse your database schemas, work with database objects and insert data as well as design and run SQL queries.

[Browse Documentation >](#)

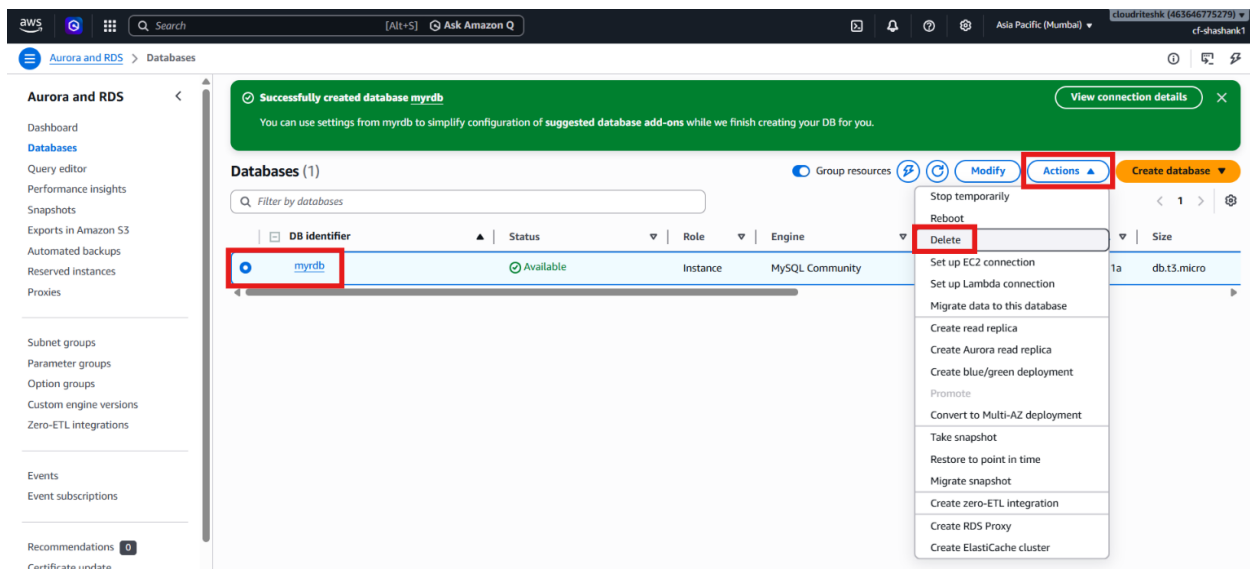


[Join the Forums >](#)

- Open the database connection successfully.
- Create a new database from **MySQL Workbench**.
- Verify that the new database appears in the schema list.



- Understand that database creation in the client is separate from RDS instance creation.
- Note that tables and schemas can now be created inside the new database.
- Return to the AWS console after testing the connection.
- Select the RDS database instance.
- Open **Actions**.
- Choose **Delete** to remove the lab database when finished.



- Type **delete me** when prompted.
- Confirm that automatic backup is disabled if required by the deletion flow.

Delete myrddb instance ✕

Permanently delete **myrddb** DB instance. You can't undo this action.

Proceeding with this action will delete the instance with all its content and can affect related resources. [Learn more](#)

☐ Create final snapshot
Determines whether a final DB Snapshot is created before the DB instance is deleted.

☐ Retain automated backups
Determines whether retaining automated backups for 1 day after deletion

☒ I acknowledge that upon instance deletion, automated backups, including system snapshots and point-in-time recovery, will no longer be available.

To avoid accidental deletion provide additional written consent.

To confirm deletion, type *delete me* into the field.

We strongly recommend taking a final snapshot before instance deletion since after your instance is deleted, automated backups will no longer be available.

[Cancel](#) [Delete](#)

- Finalize the deletion request.

Verification

- The **Amazon RDS** MySQL instance was created successfully.
- The instance reached the **Available** state.
- **MySQL Workbench** connected to the database using the RDS endpoint.
- A new database was created from the SQL client.
- The lab database was ready for cleanup after the demonstration.